

COLLECTOR EMPIRICAL OCCUPANCY VS SCENARIO LEDGER SUPPORT

Why Low Validation Loss on Held-Out Demonstrations Conceals Critical Operational Blind Spots

COLLECTOR A: CAUTIOUS TELEOP



Total Recorded Transitions:

1,000,000 samples @ 100 Hz

Scenario Grid Coverage:

8 / 600 cells (1.3%)

Zero-Count Blind Spots:

592 / 600 cells (98.7%)

Held-Out Validation Loss:

$L_{val} = 0.001$ rad (Deceptive)

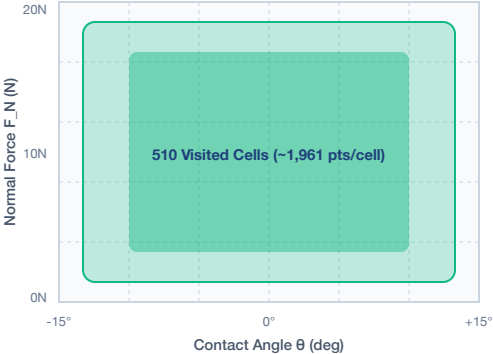
Compounding Drift Risk:

CATASTROPHIC at $\pm 2.5^\circ$

Ingestion Policy Decision:

REFUSE / REQUIRE DAgger

COLLECTOR B: EXPLORATORY POLICY



Total Recorded Transitions:

1,000,000 samples @ 100 Hz

Scenario Grid Coverage:

510 / 600 cells (85.0%)

Unmonitored Edge Cells:

90 / 600 cells (15.0%)

Held-Out Validation Loss:

$L_{val} = 0.012$ rad (Realistic)

Compounding Drift Risk:

BOUNDED by recovery data

Ingestion Policy Decision:

ADMIT / PROCEED TO TRAINING

SCENARIO LEDGER RELATIONAL JOIN

External Specification S_{req} :

Operational Envelope Grid: 600 Bins

$\theta \in [-15^\circ, +15^\circ]$, $F_N \in [0, 20 \text{ N}]$, $\mu \in [0.1, 0.4]$



Empirical Dataset Visitation: N_{cell}

Count occurrences per discretized scenario cell



Support Classification Rules:

● $N_{cell} \geq 500$: Supported Nominal Regime

Policy trained under dense supervision.

Admit directly into training cluster.

● $0 < N_{cell} < 50$: High-Variance Regime

Insufficient sample count; wide epistemic variance.

Condition: require targeted recollection.

● $N_{cell} = 0$: Zero-Support Blind Spot

Zero empirical constraints from training data.

Refuse: trigger autonomous safety gate.